PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Winpack Microsoft Intune – Tata Advanced Systems Limited (TASL)

AN INTERNSHIP REPORT

Submitted by

SYED SAIFULLA H- 20231CCS3004

Under the guidance of,

Dr. SRABANA PRAMANIK

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "MICROSOFT INTUNE IN SECURING DEVICES: BALANCING SECURITY FEATURES AND USER EXPERIENCE IN ENTERPRISE ENVIRONMENTS" is a bonafide work of "Syed Saifulla H (20231CCS3004) who have successfully carried out the Internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE ENGINEERING, CYBER SECURITY during 2025-26.

Dr Srabana Pramanik
Associate Professor
Internship Guide
PSCS (Spl. Division)
Presidency University

Dr Geetha A
Associate Professor
Program Internship Coordinator
PSCS (Spl. Division)
Presidency University

Dr. Geetha A
Associate Professor
School Internship Coordinator
PSCS (Spl. Division)
Presidency University

Dr. Anandaraj S.P
Professor & Head of the Department
PSCS (Spl. Division)
Presidency University

Dr. Shakkeera L
Associate Dean
PSCS (Spl. Division)
Presidency University

Sl.No	Name of the Examiner	Signature

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

Iam the student of final year B.Tech in COMPUTER SCIENCE ENGINEERING, CYBER SECURITY at Presidency University, Bengaluru, named Syed Saifulla H, hereby declare that the project work titled **"MICROSOFT INTUNE IN SECURING DEVICES: BALANCING SECURITY FEATURES AND USER EXPERIENCE IN ENTERPRISE ENVIRONMENTS"** who has successfully carried out the project work at **Tata Advanced Systems Limited (TASL)** and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (CYBER SECURITY) during the academic year of 2026-27. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other Institution.

Syed Saifulla H 20231CCS3004

PLACE: BENGALURU

DATE: 20-April-2026

INTERNSHIP COMPLETION CERTIFICATE

06-April-2026

TRAINING COMPLETION CERTIFICATE

This is to certify that Mr. Syed Saifulla H studying in Presidency University Bangalore successfully completed his Project Internship at Tata Advanced Systems Limited,42–43 Electronics City Hosur Road Bengaluru 560100 India, from 05-Jan- 2026 to 06-Apr-2026 under guidance of Mr. Satya Prasad

He has done his internship project on " Microsoft Intune MDM (End Point Security) & Vulnerability Assessment and Penetration Testing (VAPT) "

During his stay with us, he was found to be sincere and hardworking.

We wish him all the best for his future endeavors.

For **Tata Advanced Systems Limited,**

Prema.N

Authorized Signatory
Human Resource

TATA ADVANCED SYSTEMS LIMITED
42 - 43 Electronics City Phase - I Hosur Road Bengaluru 560 100 Karnataka India
Tel +91 80 6785 9900 / 10 Fax +91 80 6785 9901 / 02 Website www.tataadvancedsystems.com Email email@tataadvancedsystems.com CIN: U72900TG2006PLC077939
Registered Office Hardware Park Plot No. 21 Sy No. 1/1 Imarat Kancha Raviryala Village Maheshwaram Mandal Hyderabad 501 218 Telangana India

ACKNOWLEDGEMENT

For completing this project work, We have received the support and the guidance from many people whom we would like to mention with deep sense of gratitude and indebtedness. We extend our gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the project.

I am deeply thankful to **Tata Advanced Systems Limited (TASL)** for providing me the opportunity to work on this project within their esteemed organization. I extend my heartfelt thanks to the IT and security teams at TASL for their support, access to infrastructure, and real-world insights that significantly contributed to the success of this research.

I would like to sincerely thank my internal guide DR. Srabana Pramanik , Presidency School of Computer Science and Engineering, (Spl. Division) Presidency University, for her moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

i am also thankful to **Dr. Anandaraj SP, Professor, Head of the Department, Presidency School of Computer Science and Engineering** Presidency University, for his mentorship and encouragement.

I express my cordial thanks to Dr. Shakkeera L, Associate Dean, Presidency School of computer Science and Engineering (Spl. Division), Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

We are grateful to **Dr. Geetha A, PSCS** (Spl. Division) School Internship Coordinator**, Dr Geetha A, PSCS (Spl. Division) Program Internship Coordinator**, Presidency School of Computer Science and Engineering, or facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

SYED SAIFULLA H

Abstract

The proliferation of cloud computing and mobile technologies has fundamentally transformed the modern enterprise landscape, enabling unprecedented levels of workforce flexibility and operational efficiency. However, this digital transformation has simultaneously introduced significant security vulnerabilities, as organizations struggle to protect sensitive corporate data across an increasingly diverse and distributed device ecosystem. The challenge is further compounded by the need to maintain seamless user productivity and satisfaction, creating a critical tension between security enforcement and user experience. Microsoft Intune, a cloud-native enterprise mobility management (EMM) solution, provides organizations with a comprehensive framework for managing and securing devices, applications, and data across multiple platforms including iOS, Android, Windows, and macOS. Despite its widespread adoption, the actual effectiveness of Intune's security features and their real-world impact on end-user experience remains insufficiently quantified in existing literature.

Quantitative analysis revealed statistically significant variations in user impact across different security controls. Device encryption, which operates silently in the background, exhibited the lowest perceived user disruption with a mean impact score of 0.57 on a 0–2 scale. Conversely, multi-factor authentication demonstrated the highest mean impact of 1.57, reflecting the cumulative friction introduced by repeated authentication prompts during daily workflows. Remote wipe functionality showed the greatest variability in impact (standard deviation 1.07), indicating that user perception is heavily influenced by contextual factors such as device ownership and the circumstances under which remote wipe is invoked. Application protection policies yielded the most consistent impact (standard deviation 0.49), suggesting that users experience uniform restrictions across their managed applications.

Beyond the statistical analysis, this research also developed and validated a proof-of-concept integration between Intune and physical access control systems (WinPak) at the participating organization, enabling automated, context-aware policy enforcement based on employee presence. This integration demonstrated that security policies can be dynamically applied and removed in response to real-world physical events, reducing manual IT intervention while maintaining robust security postures.

The findings contribute both academic and practical insights by providing a quantitative trade-off framework that enables security administrators to make evidence-based decisions when configuring Intune policies. The framework allows organizations to prioritize high-impact features only where risk exposure justifies the usability cost, thereby achieving an optimal balance between security effectiveness and user experience. This research underscores that successful enterprise mobility management requires not only technical proficiency but also a deep understanding of human factors and organizational context.

TABLE OF CONTENT

Sl. No.	TITLE	Page No.
	Declaration	III
	Acknowledgement	IV
	Abstract	V
	List of Figures	VIII
	List of Tables	IX
	Abbreviations	X- VI
1.	**Introduction**	
	1.1 Background	
	1.2 Statistics of project	
	1.3 Prior existing technologies	01-08
	1.4 Proposed approach	
	1.5 Objectives	
	1.6 SDGs	
2.	**Literature review**	09-15
3.	**Methodology**	16-25
4.	**Project management**	
	4.1 comprehensive Project Timeline	
	4.2 Team Roles and Responsibilities	
	4.3 Comprehensive Risk Management	
	4.4 Resource Allocation and Management	26-34
	4.5 Progress Monitoring and Communication Framework	
	4.6 Challenges and Strategic Resolutions	
	4.7 Timeline Visualization and Progress Tracking	
	4.8 Future Management Considerations	
5.	**Analysis and Design**	
	5.1 Requirements	35-47
	5.2 Block Diagram	

5.3 System Flow Chart
5.4 Choosing devices
5.5 Designing units
5.6 Standards
5.7 Domain model specification
5.8 Communication model
5.9 Functional View
5.10 Operational view
5.11 Other Design Aspects

6. Hardware, Software and Simulation 48-52
6.1 Hardware
6.2 Software development tools
6.3 Software code
6.4 Simulation

7. Evaluation and Results 53-60
7.1 Test points
7.2 Test plan
7.3 Test
7.4 Insights

8. Social, Legal, Ethical, Sustainability and Safety Aspects 61-65
8.1 Social aspects
8.2 Legal aspects
8.3 Ethical aspects
8.4 Sustainability aspects
8.5 Safety aspects

9. Conclusion 66-67
References 68-70
Base Paper 71
Appendix 72-79

List of Figures

Figure ID	Figure Caption	Page No.
Fig 1.1	Sustainable Development Goals	8
Fig 3.1	System Architecture Block Diagram	22
Fig 3.3	Summary of Project Breakdown to task	25
Fig 4.1	Gant Chart	33
Fig 5.2	Functional Block Diagram	38
Fig 5.3	System Flow Chart for Automated Policy Enforcement	39
Fig 5.7	Domain Model Specification	43
Fig 5.8	Communication Model – Request-Response	44
Fig 5.9	Functional View	45
Fig 5.10	Operational View	46
Fig 7.1	System Workflow Diagram Including Key Test Points	53
Fig 7.2	Overall System Performance Metrics	58
Fig 7.3	Confusion Matrix for Profile Classification	58
Fig A	Publications & Turnitin Similarity Report	70-71
Fig A	User Home Page	75
Fig B	Intune Dashboard	75
Fig C	IOS Enrollment and Compliance	76
Fig D	Android Enrollment and Compliance	76
Fig E	Intune All Users and Audit Logs	77
Fig F	All Apps	77
Fig G	Endpoint Security	78

List of Tables

Table ID	Table Caption	Page No.
Table 2.1	Summary of Literature Reviews	13-14
Table 5.1	Summarizing requirements	35-36
Table 5.2	Comparing features of different web stacks	40-41
Table 5.3	Security Standards Applied	42-43
Table 7.1	Component Performance Metrics	56
Table 7.2	Error and Failure Analysis	57
Table 7.3	User Experience Survey Results (n = 50)	57-58

Abbreviations

Abbreviation	Full Form
AEHMS	Automated Equipment Health Monitoring System
AFV	Armoured Fighting Vehicle
API	Application Programming Interface
ATGM	Anti Tank Guided Missile
AUC-ROC	Area Under Curve - Receiver Operating Characteristic
BMCS	Bi-Modular Charge System
COTS	Commercial Off-The-Shelf
CSV	Comma-Separated Values
DHT	Digital Humidity and Temperature
DRDO	Defence Research and Development Organisation
EME	Electronics and Mechanical Engineers
ERP	Enterprise Resource Planning
FPA	Focal Plane Array
GPIO	General Purpose Input/Output
GPS	Global Positioning System
HAA	High Altitude Area
HTML	HyperText Markup Language
IFF	Identification of Friend or Foe
InfluxDB	Time-Series Database
IoT	Internet of Things
IRNSS	Indian Regional Navigation Satellite System
JSON	JavaScript Object Notation
LAC	Line of Actual Control
LTE	Long Term Evolution
MAWS	Missile Approach Warning System
MCCS	Mobile Cellular Communication System

MGB	Main Gear Box
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
MTBF	Mean Time Between Failures
NVD	Night Vision Device
OEM	Original Equipment Manufacturer
PESTEL	Political, Economic, Social, Technological, Environmental, Legal
QoS	Quality of Service
RBAC	Role-Based Access Control
REST	Representational State Transfer
RF	Random Forest
RH	Relative Humidity
ROI	Return on Investment
SAM	Surface to Air Missile
SAR	Synthetic Aperture Radar
SDG	Sustainable Development Goal
SDK	Software Development Kit
SQLite	Structured Query Language Lite
SSL	Secure Sockets Layer
SVM	Support Vector Machine
TBA	Tactical Battle Area
TI	Thermal Imaging
TLS	Transport Layer Security
UAV	Unmanned Aerial Vehicle
URL	Uniform Resource Locator
VPN	Virtual Private Network
XML	Extensible Markup Language